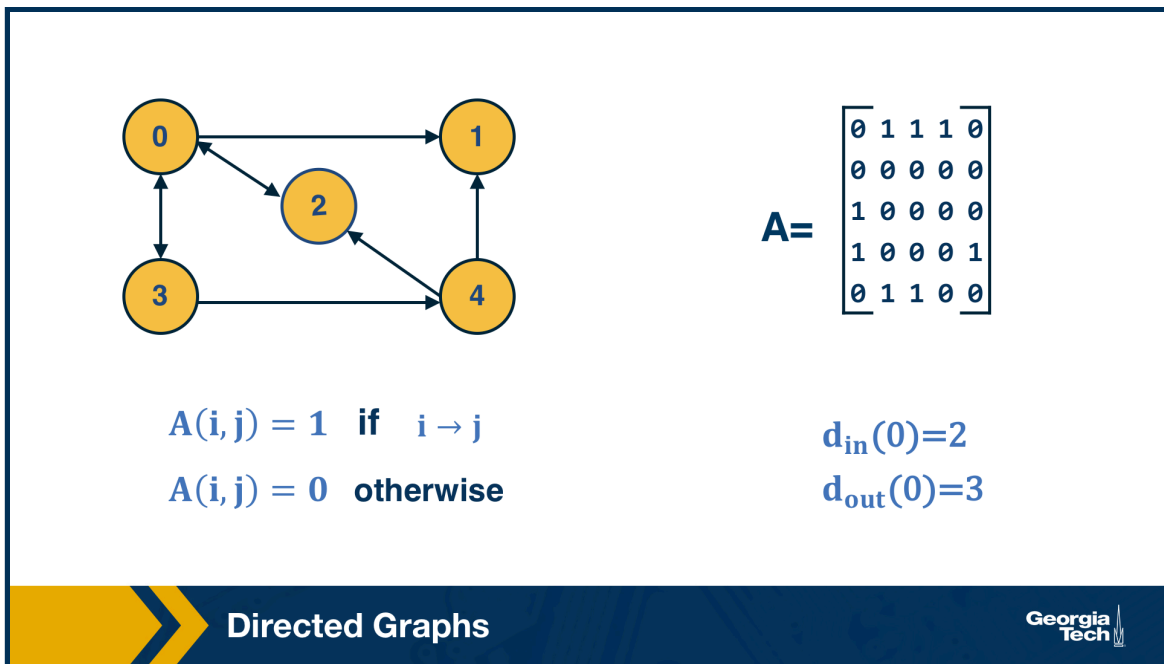


L2: Directed Graphs



Another common class of networks is directed graphs. Here, each edge (u, v) has a starting node u and an ending node v . This means that the corresponding adjacency matrix may no longer be symmetric.

A common convention is that the element (i, j) of the adjacency matrix is equal to 1 if the edge is from node i to node j – please be aware however that this convention is not universal.

We also need to revise our definition of node degree: the number of incoming connections to a node v is referred to as **in-degree** of v , and the number of outgoing connections as **out-degree** of v .

Food for Thought

Do you see that the sum of in-degrees across all nodes v is equal to the number of edges m ? The same is true for the sum of out-degrees.